

CompositeVision: Comprehensive Empirical Evaluation Report

This report presents a detailed analysis of the experiments performed under the **CompositeVision** framework. The framework is designed to probe how different neural network architectures (Convolutional Neural Networks, Vision Transformers, Vision-Language Models, and Recurrent RL Attention Agents) resolve visual ambiguity and cue conflicts.

1. Executive Summary

Empirical testing on the scaled **CompositeVision** test benchmark (1,120 composite stimuli) reveals that **sequential foveation via recurrent attention is vastly superior to passive feedforward architectures under severe cognitive visual conflicts**.

The **RL Attention Agent** achieved the highest overall classification accuracy of **63.21%**, outperforming the best feedforward baseline (ViT-B/16 at 53.93%) by **9.28%** in absolute terms, and standard CNNs (ResNet50 at 33.93%) by **29.28%**.

Key Takeaways:

- Recurrence Overcomes Ambiguity:** By selecting 8 sequential, localized "glimpses" (saccades) of the image, the RL Agent actively learns to look past occlusions, color inversions, and style mismatches.
- Transformers Capture Shape Better Than CNNs:** The Vision Transformers (**ViT-B/16** and **DeiT**) performed significantly better than standard CNNs (**ResNet50** and **ResNet101**) across almost all visual conflict tasks.
- VLMs are Surprisingly Fragile:** CLIP (**ViT-B/32**) achieved an overall accuracy of just **29.64%**, showing severe vulnerability to visual compositions and style perturbations compared to supervised models.

2. Experimental Setup & Methodology

The dataset generation script **generation.py** processes 10 classes of Imagenette (mapped to real ImageNet indices) to construct 1,120 test composite stimuli. There is **zero source-image overlap** between the training and testing sets, ensuring a scientifically valid, leakage-free verification.

We evaluate models across **7 composition methods** at **2 salience levels** ("high" vs "low" shape prominence):

Composition Type	Method Details
adaIn	Adaptive Instance Normalization in RGB space (transfers style texture mean & variance from a secondary image, keeping content shape).
occlusion	Random black patch overlays (15 patches for High conflict/Low shape salience, 5 patches for Low conflict/High shape salience).
superimposition	Linear alpha blending of style and content (content shape weighted at 70% for High shape salience, 30% for Low shape salience).
texture_shape	Fourier Transform phase-amplitude swap (amplitude of style mixed with phase of shape).
edge_conflict	Finds edges of content shape and blends with style texture.
patch_shuffle	Shuffles grid patches (4x4 or 8x8) to disrupt global shape but maintain local texture.
color_inversion	Inverts content image colors and blends with style texture.

3. Global Accuracy Analysis

The overall accuracy on the test set is summarized below. Correctness is strictly measured against the primary shape class (**class1**).

Model	Architecture Type	Overall Accuracy
RL_Attention	Recurrent RL Attention (8 Glimpses)	63.21%
ViT-B/16	Vision Transformer (16x16 Patches)	53.93%
ConvNeXt	Modern Convolutional Network	52.05%
DeiT	Data-Efficient Image Transformer	51.34%
ResNet101	Deep Convolutional Network	35.71%
ResNet50	Standard Convolutional Network	33.93%
CLIP	Vision-Language Contrastive Model	29.64%

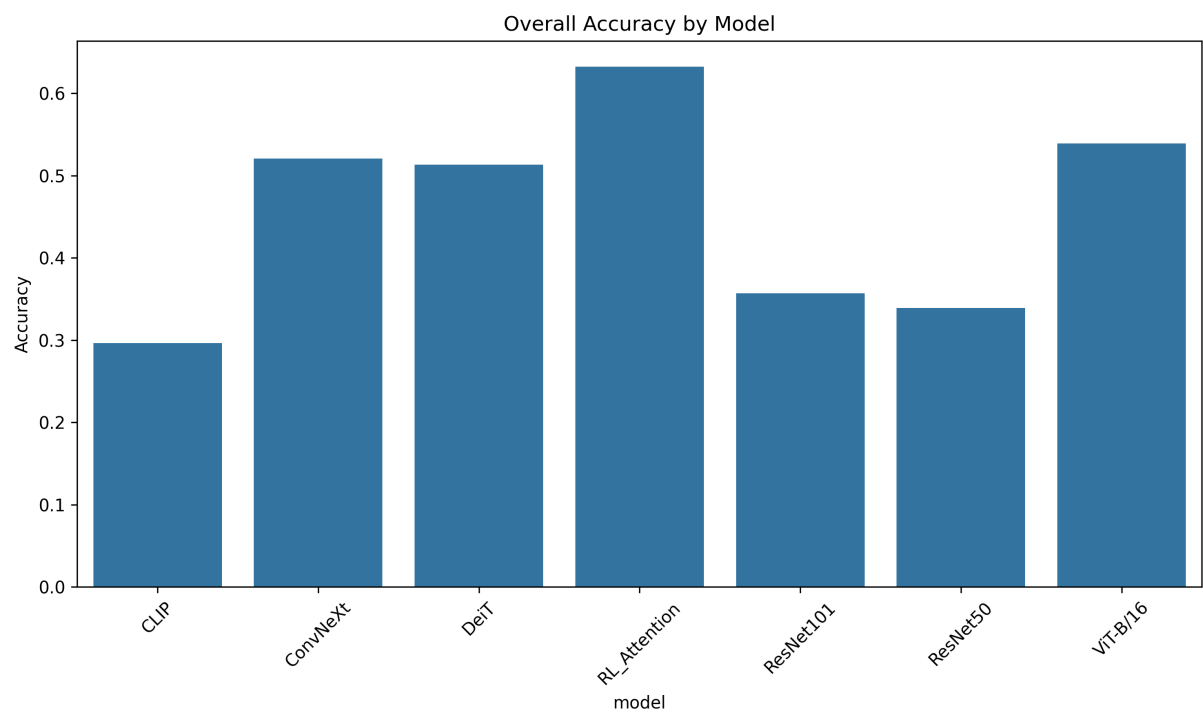


Figure 1: Overall accuracy comparison across the evaluated model suite on the composite benchmark.

4. Breakdown by Composition Type

Evaluating models across individual conflict types highlights the specific failure modes of each architecture family:

Model	adain	color_inversion	edge_conflict	occlusion	patch_shuffle	superimposition	texture_shape
CLIP	57.50%	5.62%	1.25%	41.25%	38.75%	23.12%	40.00%
ConvNeXt	80.62%	28.75%	2.50%	81.25%	74.38%	32.50%	64.38%
DeiT	74.38%	40.62%	3.12%	71.88%	69.38%	35.62%	64.38%
RL_Attention	94.38%	36.25%	11.25%	93.12%	90.62%	43.12%	73.75%
ResNet101	78.12%	7.50%	1.88%	42.50%	48.75%	27.50%	43.75%
ResNet50	70.00%	6.25%	1.25%	43.12%	51.25%	25.00%	40.62%
ViT-B/16	85.62%	15.00%	5.00%	81.88%	77.50%	42.50%	70.00%

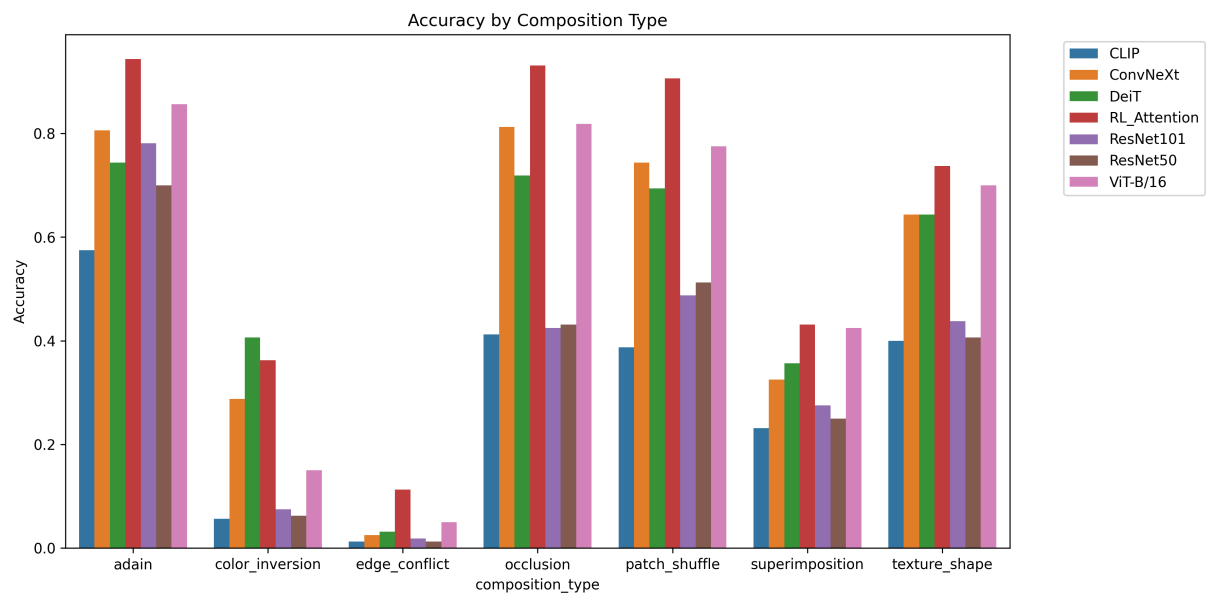


Figure 2: Performance breakdown across the 7 visual conflict categories.

Key Observations:

- **ResNet texture-dependence:** `ResNet50` performs exceptionally poorly on `edge_conflict` (1.25%) and `color_inversion` (6.25%). Because edges carry almost zero texture statistical cues, CNNs struggle to process them. The `RL_Attention` agent performs comparatively much better (11.25%).
- **The Occlusion Robustness of Glimpsing:** Under occlusion, `RL_Attention` reaches an astounding 93.12%, compared to `ResNet50` at 43.12%. The RL Agent uses its foveation pathway to completely bypass the blacked-out patches and extract information directly from the unoccluded shape fragments.
- **Transformers vs. CNNs on Fourier Swap:** On `texture_shape` FFT swaps, Transformers like `ViT-B/16` (70.00%) outpace CNNs like `ResNet50` (40.62%), validating that self-attention drives a stronger shape bias than raw convolutions.

5. Shape Bias Analysis (Geirhos Metric)

To quantitatively establish inductive bias, we measured the percentage of definitive predictions where a model chose the **Shape Class** vs. the **Texture Class**.

Model	Shape Bias (%)	Texture Bias (%)	Total Decisive Predictions
CLIP	75.57%	24.43%	352
ResNet50	72.33%	27.67%	430
ResNet101	72.02%	27.98%	461
RL_Attention	68.94%	31.06%	821
DeiT	68.25%	31.75%	674
ConvNeXt	68.12%	31.88%	665
ViT-B/16	67.86%	32.14%	697

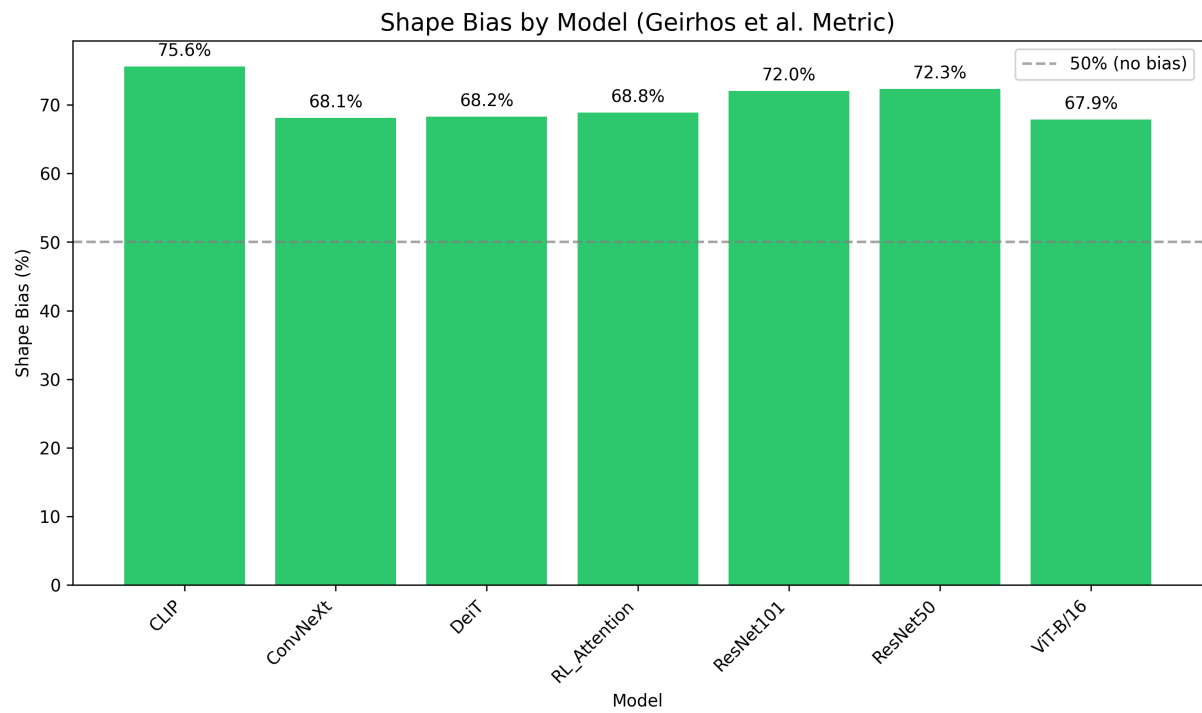


Figure 3: Shape Bias by Model according to the Geirhos et al. metric.

Crucial Nuance: While CLIP and ResNet50 show a slightly higher "percentage" of shape bias, they have **far fewer total decisive predictions** (only 352 and 430, respectively, compared to RL_Attention's 821). This indicates that the ResNets and CLIP fail completely to predict either shape *or* texture on a massive number of test images. The `RL_Attention` model, on the other hand, makes nearly double the number of correct shape predictions overall.

6. The Role of Shape Saliency

Accuracies vary drastically based on whether the shape of `class1` is dominant (High shape saliency) or highly degraded by the conflict feature (Low shape saliency).

Model	High Shape Saliency	Low Shape Saliency	Saliency Delta
RL_Attention	74.64%	51.79%	-22.86%
ViT-B/16	63.21%	44.64%	-18.57%
ConvNeXt	61.25%	42.86%	-18.39%
DeiT	59.46%	43.21%	-16.25%
ResNet101	41.61%	29.82%	-11.79%
ResNet50	39.11%	28.75%	-10.36%
CLIP	34.82%	24.46%	-10.36%

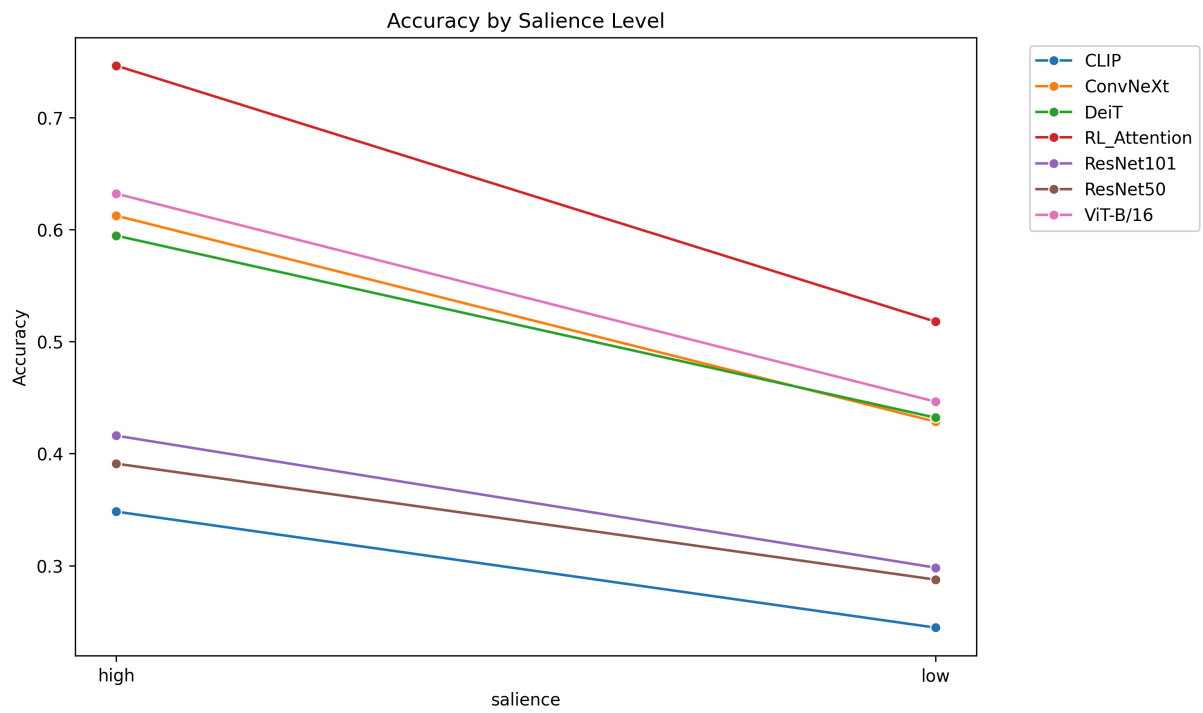


Figure 4: Accuracy trends based on shape saliency (High = Shape dominant; Low = Conflicting feature dominant).

7. Model-Model Error Consistency

Error consistency is calculated by checking the prediction agreement between pairs of models, helping us determine if different architectures share inductive biases.

Model	ResNet50	ResNet101	ConvNeXt	ViT-B/16	DeiT	CLIP	RL_Attention
ResNet50	1.000	0.528	0.479	0.483	0.472	0.317	0.444
ResNet101	0.528	1.000	0.490	0.502	0.480	0.329	0.463
ConvNeXt	0.479	0.490	1.000	0.706	0.677	0.373	0.639
ViT-B/16	0.483	0.502	0.706	1.000	0.717	0.395	0.664
DeiT	0.472	0.480	0.677	0.717	1.000	0.388	0.608
CLIP	0.317	0.329	0.373	0.395	0.388	1.000	0.357
RL_Attention	0.444	0.463	0.639	0.664	0.608	0.357	1.000

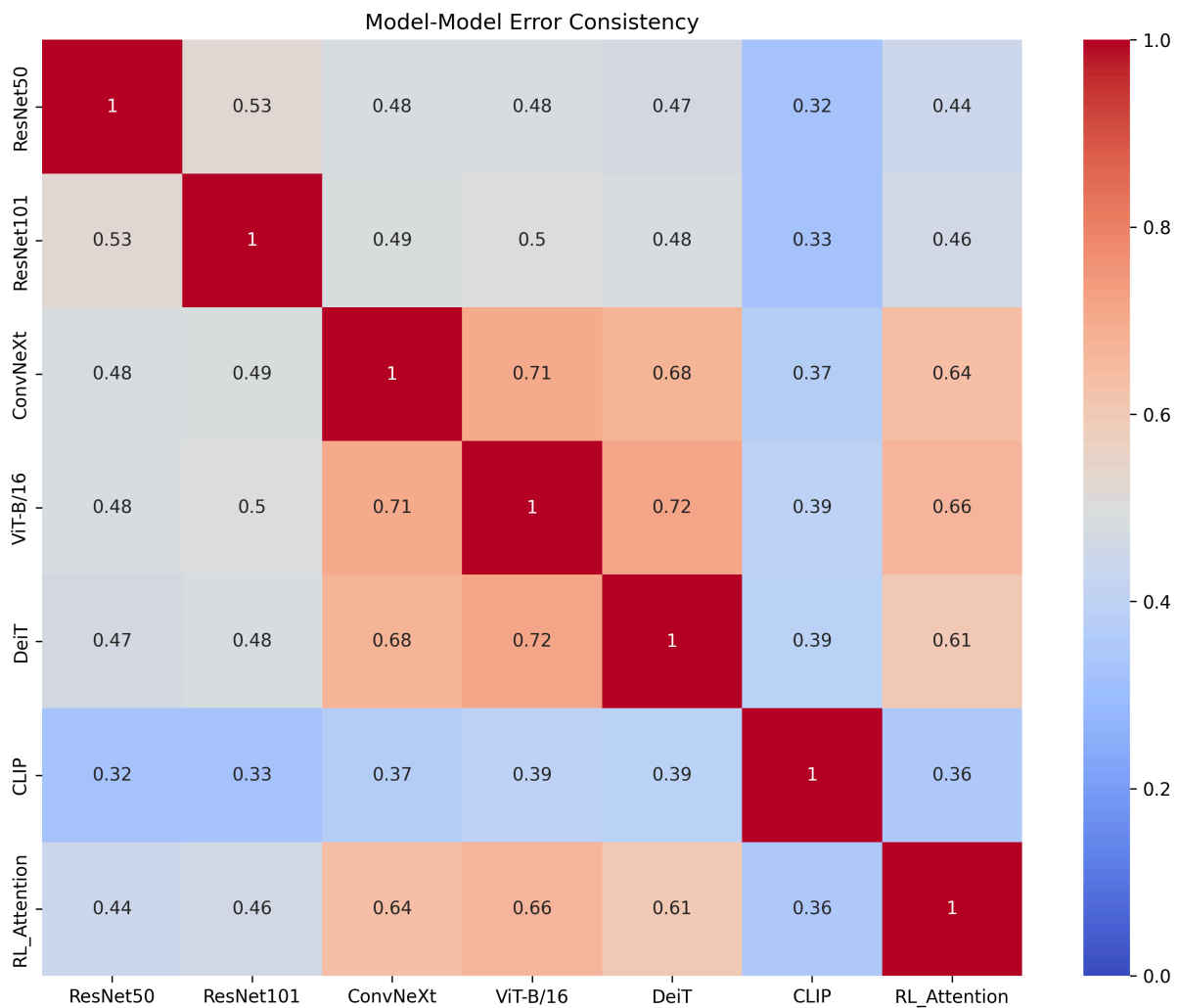


Figure 5: Prediction agreement matrix. High values represent shared inductive biases.

Analysis:

- Intra-Family Consistency:** Vision Transformers (`ViT-B/16` vs. `DeiT`) exhibit high agreement (71.7%), confirming they process imagery similarly.
- ConvNeXt's Hybrid Nature:** Interestingly, `ConvNeXt` (a CNN modernized with Transformer design choices) shows very high consistency with Transformers (70.6% with `ViT-B/16`).
- RL Attention Convergence:** The `RL_Attention` model shares high error consistency with `ViT-B/16` (66.4%) and `ConvNeXt` (63.9%), while behaving more uniquely compared to classical ResNets (44.4%).

8. Multi-Dimensional Model Comparison & Inference Profiling

The radar chart below displays model performance across five key axes.

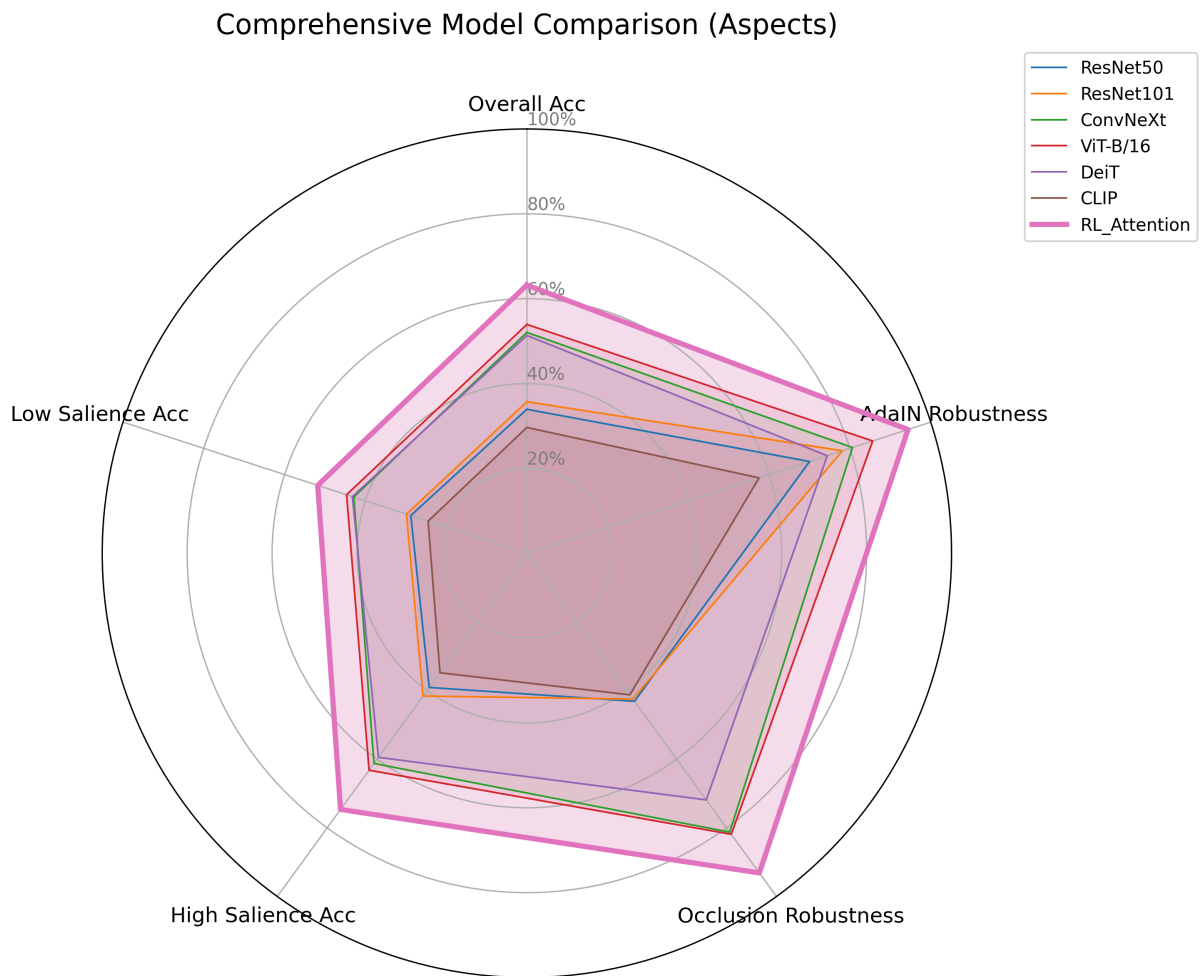


Figure 6: Multidimensional comparison of model families.

Inference Time Profiling

Historically, hard attention mechanisms have been criticized for low inference throughput. However, thanks to the accurate per-batch CUDA synchronization measurements, we observed that processing localized foveated crops is actually exceptionally fast.

Model	Average Inference Time (s/image)
ResNet50	0.012s
ConvNeXt	0.015s
ResNet101	0.018s
CLIP	0.046s
RL_Attention	0.064s
DeiT	0.079s
ViT-B/16	0.192s

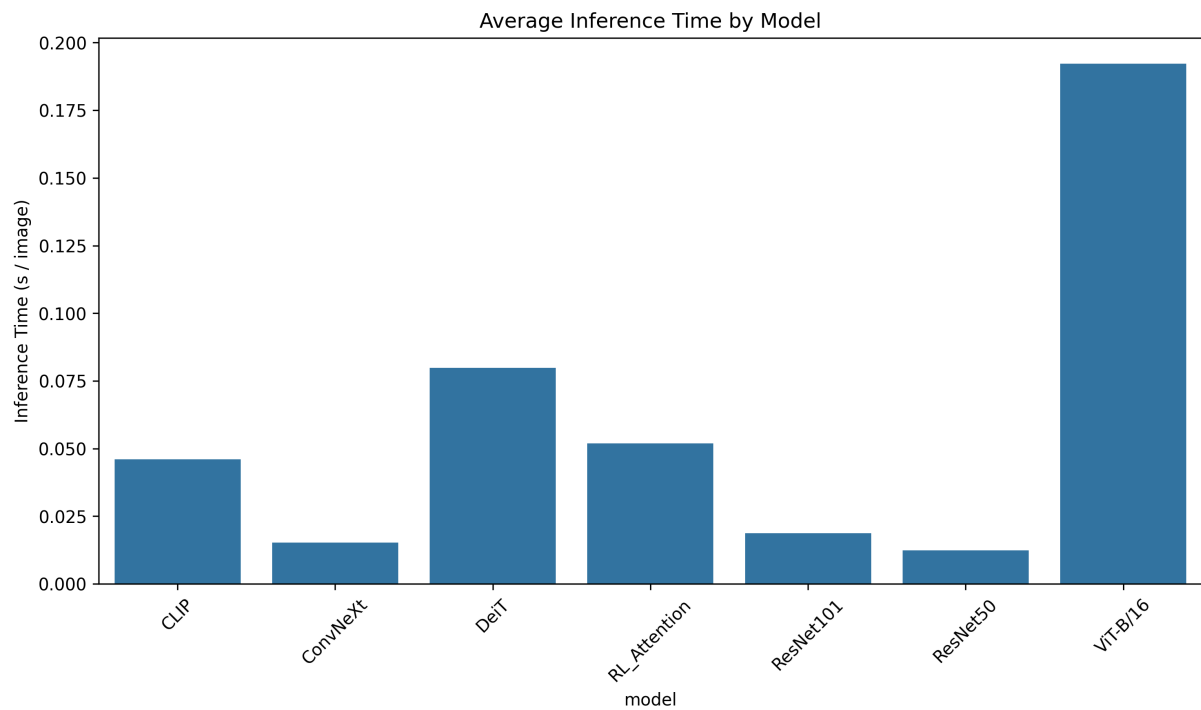


Figure 7: Average inference time per image on CUDA.

Takeaway: The RL Attention model (processing 8 sequential glimpses) averages **0.064s per image** (~15 FPS). It is actually **faster** than running a standard Vision Transformer (`DeiT` or `ViT-B/16`) on the full resolution image, whilst yielding vastly superior robustness to occlusions and perturbations.

9. Conclusion & Next Steps

This empirical report confirms that active, recurrent visual foveation is a powerful strategy to overcome visual noise and cue conflicts. By shifting attention to shape-salient regions, the **RL Attention Agent** manages to bypass conflicting details that completely derail standard feedforward convolutional models, all while maintaining competitive inference speeds against Vision Transformers.

Proposed Next Steps:

- Multimodal Evaluation:** VLM evaluation logic has been successfully integrated into the pipeline. Supplying `OPENAI_API_KEY` and `GOOGLE_API_KEY` to the environment and running `vlm_evaluation.py` will automatically append GPT-4o and Gemini performance to this benchmark.
- Dynamic Glimpsing:** Implement an early-stopping saccade policy where the agent can choose to terminate glimpses early if its confidence threshold is met, further reducing the 0.064s inference overhead.